

UMANG SHAH

Data Engineer – New York

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Summary

Data Engineer with 4+ years building scalable ETL, real-time streaming, & cloud-native platforms across fintech & healthcare.

Education

New York University – M.S. in Computer Science, GPA: 3.96/4.0

Sep 2022 – May 2024

VIT Chennai – B.Tech in Computer Science and Engineering, GPA: 9.16/10

Jul 2015 – May 2019

Experience

Analyst (Data / Software Engineering), Nomura Holdings America – New York, NY

Jul 2024 – Present

- Delivered **60+ core features** on an enterprise .NET (C#) pricing platform by modernizing pricing logic & trade workflows.
- Designed and prototyped a real-time **Spark Streaming** pipeline to track trade volume and revenue across multiple dimensions, ensuring reliability with **fault-tolerant features** such as checkpointing and watermarking.
- Implemented a **fast estimation feature** to provide instant pricing approximations for incoming trades, enabling quicker decision-making before final calculations.
- Implemented **automated pipelines** to replace Excel-based processes, reducing manual effort for traders.
- Automated **post-trade workflows**, including trade ingestion, data creation, position tracking, and contract generation, allowing traders and sales teams to complete compliance checks faster and more reliably.
- Implemented trade booking across internal & external systems (e.g., **Radar, Bloomberg APIs**) enabling risk management & trade visibility.

Software Engineer Intern, MIO Partners (McKinsey Investment Office) – New York, NY

Jun 2023 – Aug 2023

- Migrated ETL workflows from **ActiveBatch + EBS** to **Step Functions + S3**, improving reliability & reducing cost.

Senior Software Engineer (Data Engineering), UnitedHealth Group – Hyderabad, India

Jul 2019 – Jul 2022

- Delivered **MTD/YTD analytics pipelines** for **26M+** members on large **Hive/Spark** clusters, leveraged **PySpark** for transformations of varying complexity, optimizing cluster resources to ensure stable, high-throughput processing.
- **Reduced data SLA from 6 hours to 10 minutes** by designing and deploying **Kafka-based** streaming pipelines, enabling faster insights for business stakeholders.
- Standardized healthcare provider onboarding by building pipelines to ingest **flat files, CCD XML, HL7 messages**, and **Kafka** streams into partitioned **Hive Parquet** datasets, reducing onboarding errors and processing time.
- Improved data quality and reliability through ingestion validation, workflow orchestration, and client scheduling controls, minimizing downstream data errors and delays.
- Reduced infrastructure cost and production downtime by migrating core services to **Kubernetes**.
- Enabled monitoring & alerting with **Grafana & Prometheus**, accelerating incident detection & downtime prevention.

Skills

Data Engineering: Spark, Kafka, Hive, HDFS, Snowflake, Airflow, ETL, PostgreSQL, SQL, MongoDB, HBase, Redis, KDB+
Cloud & DevOps: AWS, Azure, Kubernetes, Docker, CI/CD, Jenkins, Nginx

Web & Backend: Node.js, Flask, .NET Framework, FastAPI

Analytics & ML: Pandas, Tableau, D3.js, Chart.js, NLTK, StanfordNLP

Programming & Version Control: Python, JavaScript, Java, C#, Git

Projects

Media Magic (Live Demo)

Sep 2023 – Dec 2023

- Designed and developed a **scalable on-demand image optimization** platform including **React** and **Python SDKs**, enabling developers to seamlessly deliver responsive, right-sized images across devices with minimal integration effort.
- Built a **Quick-Check page** that previews performance gains from any URL before SDK integration, with results across mobile and desktop viewports.
- Implemented **event-driven data pipelines** and per-image analytics tracking **cache hit ratios, resolution distributions, load time**, and transformation latency, with **real-time dashboards** and a developer overlay for inspecting image loading and optimization.

Forest Fire Tracker (GitHub)

Feb 2023 – Apr 2023

- Built a **Kafka-driven** streaming pipeline to process **Unity**-generated satellite-image grids across ingestion, preprocessing, and analytics stages.
- Generated **near-real-time** fire-risk progression states by implementing **OpenCV**-based tree/cluster detection, fire-region extraction, and iterative spread prediction.
- Delivered reliable, live operational visibility by orchestrating **Redis** state management, **Dask** aggregation, and a **Dash** metrics dashboard with flow-control signaling.

Awards

- Won “Open Gov Data Hack” 2018, a national-level competition organized by NIC, Government of India.
- Won best project award in Technical Expo’18, VIT Chennai.